



Enterprise Asset Management (EAMS) v15

Assets

End User Documentation

Update 3/1/2024

Managing Assets

Managing enterprise Assets is a core function of MaintStar EAMS. Assets are resources of which an organization is responsible for maintaining throughout its life from acquisition to retirement/salvage. Following are some key components of managing assets:

- **Asset:** the resource to be managed and maintained – this may be a vehicle, road, stormwater channel, building, generator, etc
- **Asset Type:** the category for the asset – for instance all road segments would be grouped into a “Road” Asset Type, or all light duty vehicles grouped into a “Light Truck” Asset Type
- **Asset Group:** to ease Work Order scheduling, assets may be grouped together, even across Asset Types – for instance, assets requiring monthly inspection can be grouped either by type, location or other factors
- **Bill Rate:** a cost can be assigned for an assets use – for instance vehicles can be applied a rate for when they are added to a Work Order LEm
- **Asset Customization:** screens can be customized to contain fields unique to a particular asset type
- **Condition Index:** can be applied to assets according to industry standard asset ratings
- **Asset History:** a summation of work order history and cost is available for each asset
- **Asset Parts:** parts used against assets can be tracked to show what part was used and the last time it was used
- **Asset Warranties:** assets with warranties can be tracked to provide greater awareness to technicians of covered assets
- **Asset Notes and Warnings:** notifications can be applied to assets to show a warning on the asset and/or work order, for example a vehicle may show a warning to indicate a safety condition
- **Meter Type:** assets can be assigned meter types to track mileage, rotations, counts, etc
- **Work Order Asset:** asset(s) loaded onto a work order – all work order history will be tracked
- **Work Order Meter:** as assets with meters are worked on, technicians can update meter reads
- **Consequence and Probability of Failure:** COF and POF are industry standard ratings to show the importance and criticality of maintaining assets
- **Searching for Assets**
- **Asset Reports**

Asset Types

Managing Assets requires a higher level of permission and is typically done so by system administrators.

Assets are always assigned to Asset Types. Before creating or uploading assets a corresponding asset type must exist. Asset Types can be accessed from the Asset > Asset Type menu.

The screenshot shows the MAINTSTAR Asset Management interface. On the left is a blue sidebar with a menu. The main area is titled 'Asset Type' and shows a list of asset types filtered by 'FACIL (285)'. The list includes columns for Edit, Asset Type, Description, Active, Equipment, Vehicle, Workgroup, Customization Class, Meter Type 1, and Meter Type. The table contains 10 rows of data.

☐ Edit	Asset Type	Description	Active	Equipment	Vehicle	Workgroup	Customization Class	Meter Type 1	Meter Type
☐ Edit	Air Compressor	Air Compressor	☒	☐	☐	FACIL	Boiler		
☐ Edit	Auger	36" Post hole Auger	☒	☐	☐	PARK			
☐ Edit	Basement	Basement	☒	☐	☐	FACIL	Building		
☐ Edit	Bathroom	Bathroom	☒	☐	☐	FACIL			
☐ Edit	Boiler	Boiler	☒	☐	☐	FACIL	Boiler		
☐ Edit	Building		☒	☐	☐	FACIL	Building		
☐ Edit	Car	4 door sedan	☒	☒	☒	FLEET	vehicle		
☐ Edit	Control Panel	Control Panel	☒	☐	☐	WATER			

Creating a New Asset Type

Access the Asset Type menu. This list will display all current Asset Types, or alternatively they can be filtered by Workgroup. To create a new Asset Type for one that does not currently exist, click the green “New Asset Type” button.

Provide a Asset Type name, Description and specify the assigned Workgroup.

NOTE: Asset Types can be shared across multiple Workgroups by accessing the “Filter” options in Workgroup Settings.

The screenshot shows the 'Asset Type: Air Compressor' details page. It includes a sidebar, a top navigation bar, and a main content area with tabs for Details, Attachments, Asset Subtype, Subsystem, and Condition Indexes. The 'Details' tab is active, showing fields for Asset Type, Description, Workgroup, and Auto-Key settings.

The screenshot shows the 'Create New Asset Type' dialog box. It has fields for Asset Type, Workgroup, and Description, and buttons for Add and Cancel.

Asset Type Details

Once the Asset Type is created additional attributes can be assigned. Many are advanced functions, however the following is a summary of the more commonly used fields.

- **Auto Key Pattern:** if the organization does not currently have an asset numbering scheme, this can be used to effectively start at “1” for a given asset type
- **Customization Class:** each asset can have it’s own customization screen
- **Parent Asset Type:** Asset types can be linked together in a hierarchy
- **Equipment and Vehicle Flags:** used to flag whether assets can be used as resources on a Work Order
- **Bill Rate:** Cost of using (not maintaining) an asset
- **Unit Name:** Used in conjunction with Bill Rate (hour, mile, etc)
- **Meter Type:** Used to specify the type of meter used on an asset (ie vehicle: odometer)
- **Auto Key Pattern:** When a WO is created

The screenshot shows the MAINTSTAR Asset Management interface. The sidebar on the left contains a 'Full menu' dropdown and a list of navigation items: Home (Admin), Work Order, Work Request, Asset, Asset Type (highlighted), Asset Group, Warranties, Barcode, Asset Claim, Lease, Acquisition, Disposal, Registry, Reports, Asset Management, Inventory, Reporting, Work Planning, Purchase Orders, Invoicing, Fleet, Setup, and Security. The main content area is titled 'Asset Type: Air Compressor' and includes tabs for Details, Attachments, Asset Subtype, Subsystem, Condition Indexes, and Assets. The 'Details' tab is active, showing a form with various fields for configuring the asset type. Fields include Asset Type (Air Compressor), Workgroup (FACIL), Description (Air Compressor), Auto-Key New Assets (checkbox), Auto-Key Pattern, Auto-Key Last No, Remarks, Customization Class (Boiler), Reorder Level (.000), Bill Rate (.00), Parent Asset Type, Reorder Qty (.000), Plan Rate, Equipment (checkbox), Tool (checkbox), Facility (checkbox), Vehicle (checkbox), Linear (checkbox), Depreciation Formula (10Yr-SL-Mo), Default COF (4), Unit Name, Cost Share Col, Asset Printout (186 - Asset Printout), Aging Code, Fleet/Equip Details, Downtime %, EOL Age Years (14), EOL Percent, Max MPG, Meter Increment Min, Meter Increment Max, Motor Pool (checkbox), Max Dntime Hrs, EOL Meter, Min MPG, Meter Type 1, Meter Type 2, and Meter Type 3.

Creating Assets

Most importantly, Assets must have a unique Asset ID. Asset IDs are unique to the Asset Type. This means an Asset ID of “1” can be assigned to “Trucks” and “Roads” separately, however, duplicate “1’s” can’t exist in “Trucks” Asset Types. If there is any doubt about importing or numbering Assets (to include via GIS or import functionality) coordinate with MaintStar.

To Create a New Asset, navigate to Assets > Assets and click the green “New Asset” button. As previously indicated, be sure that a corresponding Asset Type exists. Specify the Asset Type, a unique Asset ID (if “Auto Key” is not used) and provide a description.

The screenshot shows the 'Create Asset' dialog box. It has a title bar with a close button (X). The dialog contains three input fields: 'Asset Type*' (a dropdown menu), 'Asset ID*' (a text field), and 'Description' (a text field). A green 'Create' button is located at the bottom right of the dialog box.

Asset Details

As with most screens in MaintStar, Assets have their own details page on which data can be stored. Asset screens are completely customizable and can look different from examples in this manual or in the system itself.

Common Asset Details can include:

- **Description**
- **Account Number**
- **Location**
- **Ratings**
- **Attachments**
- **Purchase Data**
- **Status and Sub Status**

Activating/Deactivating Assets

Assets will always have either an “Active” or “Inactive” flag assigned to them. Furthermore, they may contain sub status information to provide additional status details, for instance, “Active – In Service” or “Active – In for Repair”. Sub Status data is stored in the “Asset Sub Status” setup table which is typically maintained by administrators.

NOTE: Most reports and filters will default to only show “Active” assets so this must be kept in mind.

Attachments

Attachments can be uploaded to assets. This includes images and documents. This can be extremely helpful for reference purposes. When a asset with an image is assigned to a Work Order, that image can be displayed on the Work Order (depending on customizations).

As with Work Order attachments, images can be assigned as thumbnail images to personalize the screen.

Asset: 0004
Type: pkLIGHTS

Details Parts Meters History Attachments >>

New Asset Save < > Create WO

Asset Id 0004 Asset Type pkLIGHTS - Lights

Asset Subtype Description

☒ Active Sub Status

Location Sub Location

Parent Asset

Notes

Rating Criticality 1

Warning

Customer ID Age yrs

Acct No ☐ GIS Verification Needed

Asset: 0
Type: Building

Details Parts History Attachments (1) Meters >>

New Asset Save < > Create WO

Alerts ~
☒ There is a warranty.

Asset Details

Asset Type Building

Asset Id 0 Parent Asset Id 001.001.010.14.080

Asset Descr Plant # 2 Water

Owner High Point Exterior Color Scheme

Asset Notes

Mfg No Sub Status In Repair

Loc Cd 007 - AMADOR THE

Asset: 0
Type: Building

Details Parts History Attachments (1) Meters >>

Add Attachment

Edit	Name	Description	Thumbnail	Print	Origin	Added By	Date
☐ Edit	Document_367.tif		☒	☐	Asset	ERIC	06/

Thumbnail image of a building.

Asset Notes

Notes or Warnings can be assigned to Assets to show as alerts on both Assets and Work Orders. This can be especially helpful when unique Notes are applied to Assets. To add a Note or Warning simply enter the data into the Asset Notes field.

NOTE: For these fields to display the screen customization must contain those fields AND Alerts and Workgroup settings must be configured.

The image contains two screenshots from the MAINTSTAR application. The left screenshot shows the 'Asset: 45178' page with 'Type: Building'. The 'Alerts' section is highlighted in pink and shows a note: 'Notes: Ferrari of Beverly Hills.' The 'Asset Notes' section is highlighted in yellow and shows the same note. The right screenshot shows the 'Work Order: 0000020191' page. The 'Alerts' section is highlighted in yellow and shows the note: 'Asset 45178 has note: Ferrari of Beverly Hills.' The 'Work Order' page also shows various fields like 'Wo No', 'Issued', 'Priority', 'Actual Start Dt', 'Activity Code', 'Wk In Progress', 'WO Summary', 'Assign To', 'Supervisor', 'Action Taken', 'Workgroup', and 'Acct No'.

Asset Parts

Assets can track part numbers, quantities and install dates. If parts are applied to a work order from inventory against an asset, MaintStar will automatically apply the parts to the asset. This can be invaluable when researching part numbers for a specific asset or even to find out when it was last used.

In the event inventory isn't used on Work Orders, parts can be manually added to the asset as a reference. To do so, go to the Asset, click on the Parts tab, click "Add". Select the part from the inventory and save.

The image shows a screenshot of the 'Asset: 45178' page with 'Type: Building'. The 'Parts' tab is selected and highlighted in blue. A yellow 'Add' button is visible in the top left corner of the Parts section.

Asset Meters

Assets equipped with meters can store meter data which is helpful in tracking an Assets use and for scheduling maintenance routines (Preventative Maintenance – PMs). In order to use meters on an Asset, the Meter Type must be configured (Setup Tables) and then applied to the Asset Type. The example below shows an Asset with a “Miles” meter (odometer).

Once the Meter Type has been applied to the Asset Type, it can then be updated on the Asset(s) itself. To do so, locate the Asset, click on the Meter tab, then update meter information with the current meter reading.

MAINTSTAR Asset Management

Full menu ▾

- Home (Admin)
- Work Order
- Work Request
- Asset
- Asset Type
- Asset Group
- Warranties
- Asset Claim
- Lease
- Acquisition
- Disposal
- Registry
- Reports
- Asset Management
 - Inventory
 - Reporting
 - Work Planning
 - Purchase Orders
 - Invoicing
 - Fleet
 - Setup
 - Security

Asset Type: Truck

Details Attachments Asset Subtype Subsystem Condition Indexes Assessments

+ New Asset Type Save < > Show Asset Count

Asset Details

Asset Type: Truck Workgroup: FLEET

Description: 1/2 ton service truck

☐ Auto-Key New Assets

Auto-Key Pattern: Auto-Key Last No:

Remarks:

Customization Class: vehicle ☒ Parent Asset Type:

Reorder Level: .000 Reorder Qty: .000

Bill Rate: .00 Plan Rate:

☒ Equipment ☐ Tool ☐ SVF Flag

☒ Facility ☐ Can Borrow / Check Out

☒ Vehicle ☐ Linear

Depreciation Formula: Cost Share Col:

Default COF: Asset Printout:

Unit Name: Aging Code:

Fleet/Equip Details

Downtime %: Max Overtime Hrs:

EOL Age Years: 10 EOL Meter:

EOL Percent: Min MPG:

Max MPG: Meter Type 1: Miles

Meter increment Min: Meter Type 2:

Meter increment Max: Meter Type 3:

MAINTSTAR Asset Management

Full menu ▾

- Home (Admin)
- Work Order
- Work Request
- Asset
- Asset Type
- Asset Group
- Warranties

Active Assets* (45)

truck List List+ Tree

Asset ID	Asset Type
223	regDUMP
225	regDUMP
226	regPATCH
227	regDUMP

Asset: 77452 Type: Truck

Details Parts Meters History Attachments (1) >>

+ Add Meter + Save + Reset

Cancel	Action	Meter Type	Initial Reading	Current Reading	Offset	LTD Reading	Rollover Date	M
Cancel	Replace Meter	Miles	2300.00	2300.00	.00			

As Work Orders are created against Assets with Meters, meter data can be recorded on the Work Order itself, then it will update the Asset once the Work Order is closed.

MAINTSTAR Work Order

Full menu ▾

- Home (Admin)
- Work Order
- Work Request
- Asset
- Asset Type
- Asset Group
- Warranties

Work Order: FL24000398

Details Assets (1) LEM (1) Assets History Job Site Status Related Attachments Log Watchers >>

+ Add Asset + Save + Reset + Add Multiple + Add Group

Cancel	Done	Compl. Date	Cost Share	Percent	Cur. Meter Read	WO Meter Read	Sub Status	VIN
Cancel	Done	03/08/2024	100.00	100.0%	6,303 (Miles)	6,303 (Miles)	Check...	9A654987321

Cost Share Total: 100 Percent Total: 100.0%

Probability/Consequence of Failure

Not all assets are equally important to a system; some assets are highly critical to achieving system's Level of Service goals and others are much less so. Criticality is the measure of risk associated with an asset. Projects and activities should be prioritized based on criticality analysis in order to use limited financial and personnel resources efficiently. To determine where to allocate resources, it is important to determine which assets are the most likely to fail and the consequences if the failure occurs.

Furthermore, the criticality of assets is system specific. Certain assets or types of assets may be critical in one location but less critical in another. Staff must examine the assets in their inventory carefully to determine which assets are critical and why.

To determine which assets are critical, two questions are important:

- How likely the asset is to fail (probability of failure)?
- What are the consequences if it does (consequence of failure)?

MaintStar has the ability to track and calculate criticality based on the POF and COF. Typically, "Criticality = POF x COF". Assets have several modes of failure beyond simple mortality and there are many factors impacting the likelihood of a given asset failing. The factors taken together provide an overall assessment of the asset's probability of failure and can be represented with a numerical rating scale (e.g., 1 to 5 or 1 to 10) ranging from a very low probability of failure to a very high probability of failure. In assessing risk, the lower number is always for lower probability and the higher number for higher probability. A zero probability is not used because no asset has virtually zero risk. There is always at least a small chance the asset could fail. After the likelihood of failure of each asset is determined, it is important to determine how bad a failure would be. This determination of consequence of failure involves the consideration of several tangible and intangible factors and can also be expressed numerically using a similar rating scale (e.g., 1 to 5 or 1 to 10) as probability of failure.

In the example to the right, the POF is 1 and the COF is 5 giving the Asset a criticality of 5.

Asset: 11528
Type: Car

Details Parts Meters History Attachments (2) »

New Asset Save ← → ↺ Create WO ⓘ

Alerts ^
 Recall on Airbags
 ✓ There is a warranty.
 Due PM: Annual Service #3

Asset Id	11528	Asset Type	Car - 4 door sedan
Model Year	2020	Make	Ford
Model No	Focus	Vin	F145785628
Lic No	E124E124	Item Color	Midnight Blue
Install Date	01/10/2019	Purchase Date	03/25/2020
Asset Descr	4 door sedan		
Date Built	12/25/2017	Original Value	25000.00
Current Value	15000.00	Replace Cost	35000.00
Fuel Type	UNL	Owner	John Franks
VEU	20000	Active	☒
Sub Status	In Service	Loc Cd	034 - OSC SUPPOF
Asset Notes			
Mfg No	Ford	Age Yrs	5
Rem. Life Years	7	Rating Value	2.00
POF	1	Useful Life Yrs	15
COF	5	Criticality	5
Salvage Value	2000.00	Date in service	

Asset History

As Work Orders are created and closed, Assets are automatically updated with this information providing research and lookup capability. In addition to Work Order details, a summation of Work Order cost is provided.

Asset: 11528
Type: Car

Details Parts Meters **History** Attachments (2) »

☐ Include Children

☐ WO Only ☐ WR Only ☒ Both

WO #	Issued ↑	Target	Closed	Compl. Date	Status	WO Cost	WO Descr
WR00000118	05/18/2022	05/23/2022			O	\$0.00	
FLE2200046	03/11/2022	03/11/2022	03/11/2022	03/11/2022	C	\$120.00	engine missing/ not r
WR00000086	03/11/2022	03/21/2022	03/11/2022		C	\$0.00	
FLE2200045	03/10/2022	03/10/2022	03/10/2022	03/10/2022	C	\$120.00	engine isn't running c
WR00000080	03/10/2022	03/20/2022	03/10/2022		C	\$0.00	
FLE2200038	01/17/2022	01/31/2022			O	\$425.92	Flat tires
FLE2000025	07/07/2020	07/07/2020	07/07/2020		C	\$42.00	replace brake pads
FLE2000024	07/06/2020	07/06/2020	07/06/2020		C	\$40.45	fix oil leak
						WO Cost Total: \$748.37	

Creating a Work Order from an Asset

Assets can be assigned a multitude of ways, from a Work Order, from the Map (if GIS integration is utilized) or from the Asset itself. To create a Work Order from the Asset, simply click the “Create WO” button from the applicable Asset.

Once the Work Order is created a dialog box will provide confirmation as well as a link to the Work Order.

Asset: 11528
Type: Car

Details Parts Meters History Attachments (2) »

New Asset Save Create WO

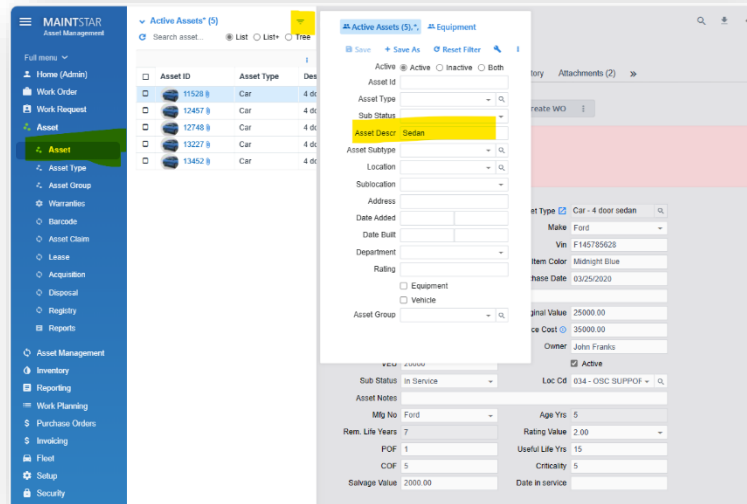
Alerts ^
Recall on Airbags
✓ There is a warranty.
Due PM: Annual Service #3

Asset Id 11528 Asset Type Car - 4 door sedan
Model Year 2020 Make Ford
Model No Focus Vin F145705620

Asset Filters and Searches

As with Work Order lookup, finding Assets can be achieved using the Asset Filter tool in the Asset > Asset list. In addition to looking up Assets, custom Asset filters can be created on an individual user basis. The example to the right shows a basic search for the word “sedan” from all assets associated to that particular Workgroup.

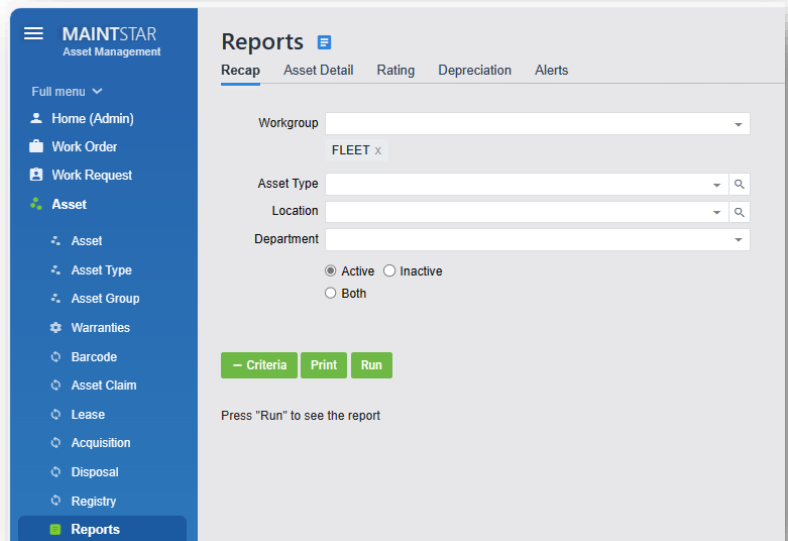
To Save the custom filter simply click “Save As”, and provide a name for the search.



Standard Asset Reports

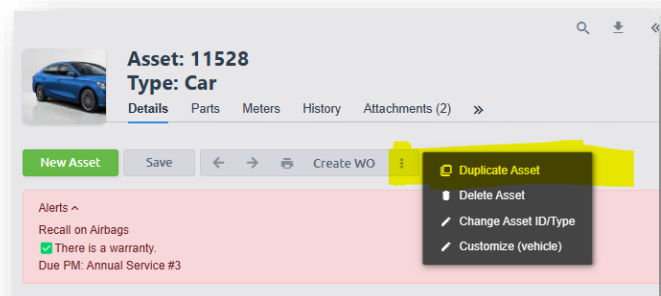
Asset Reports can be accessed from the Assets > Asset Report screen. These are basic fixed reports and are often supplemented through the use of “Reports and Queries” when special conditions are required. Most frequently used reports in this section include:

- Recap: A count of Assets by Asset Type Type, Location or Department
- Asset Detail: A detail of assets, which can be filtered and exported

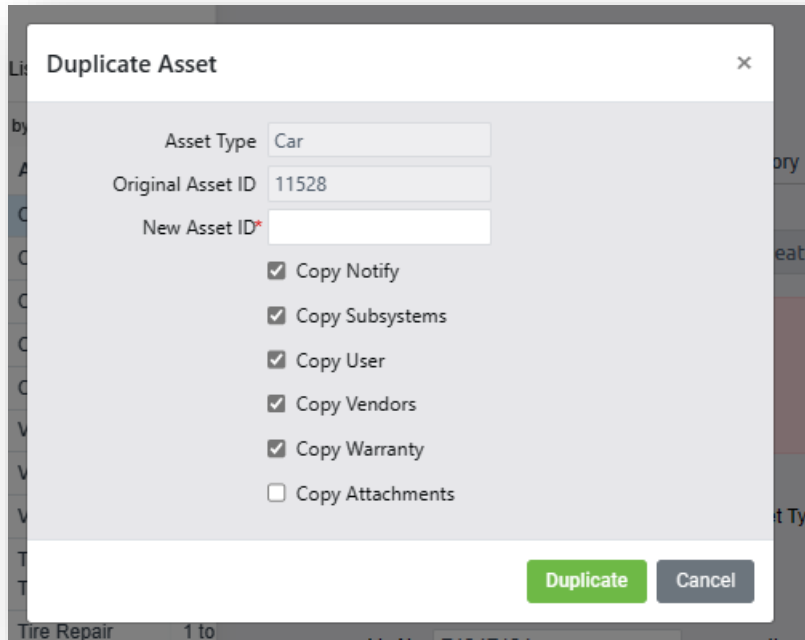


Duplicating Assets

Assets of the same Asset Types can easily be duplicated making Asset entry easier. As with most Asset changes, this is dependent on privileges assigned to users. To duplicate an asset, go to Assets > Assets, find the appropriate Asset and click the ellipses icon. When an Asset is duplicated, a new Asset ID MUST be assigned.



When the duplication process is started, a prompt will provide the ability to copy other pertinent Asset data. This is optional and helps to reduce key strokes. Once the duplication is complete update the remaining Asset data.



The 'Duplicate Asset' dialog box is shown. It contains the following fields and options:

- Asset Type: Car
- Original Asset ID: 11528
- New Asset ID*: (empty field)
- ☒ Copy Notify
- ☒ Copy Subsystems
- ☒ Copy User
- ☒ Copy Vendors
- ☒ Copy Warranty
- ☐ Copy Attachments
- Buttons: Duplicate (green), Cancel (grey)

Deleting Assets

MaintStar recommends extreme caution when deleting Assets. This feature should only be used when a mistake is made to an Asset. If an Asset is sold, salvaged, damaged beyond use, the appropriate action is to "Deactivate" the Asset. If the choice is made to delete the Asset, locate the Asset to delete, click the ellipses and select "Delete Asset".

